ExposeCheck Tutorial: FITS Image Saturation Analysis

Overview

ExposeCheck is a Python utility that analyzes FITS astronomical images to detect pixel saturation. Saturation occurs when pixels reach their maximum value (typically around 65,000 ADU for 16-bit cameras), causing loss of data and photometric accuracy. This tool helps you quickly identify which images in your observing run may have saturation issues.

Installation Requirements

Before using ExposeCheck, ensure you have the required Python packages:

```
pip install numpy astropy
```

Analysis Modes

ExposeCheck offers three analysis modes, each examining different regions of your images:

1. `-target` Mode (Inner 20%)

Analyzes the central 20% of the image (1/10 of dimensions from center in each direction). This is ideal for checking if your primary astronomical target is saturated.

Use for: Science targets like stars, galaxies, or transiting exoplanets

2. `-center` Mode (Inner 50%)

Analyzes the central 50% of the image (1/4 of dimensions from center in each direction). This larger region is useful for calibration frames where you want to check a substantial portion but avoid edge effects.

Use for: Flat fields, calibration frames, or wide-field surveys

3. `-all` Mode (Entire Image)

Analyzes the complete image. Useful when you need to check for saturation anywhere in the field.

Use for: Full-field checks, finding saturated field stars, quality control

Usage Examples

Example 1: Checking if NGC 7331 Galaxy is Saturated

```
python exposecheck.py /NGC7331 -target
```

This command will:

- Scan all FITS files in the `/NGC7331` directory
- Check only the central 20% where NGC 7331 should be centered
- Ignore potentially saturated field stars in the outer regions
- Generate `NGC7331_target.csv` with the analysis results

Example 2: Checking if Star HAT-P-31 is Saturated

```
python exposecheck.py /HAT-P-31 -target
```

For exoplanet host star observations:

- Analyzes the central region where HAT-P-31 is positioned
- Critical for ensuring your photometry target isn't saturated
- Produces `HAT-P-31_target.csv` report

Example 3: Checking Flat Field Calibration Frames

```
python exposecheck.py August_2025_flats -center
```

For flat field validation:

- Checks the central 50% of each flat frame
- Avoids vignetting effects at the edges
- Creates `August_2025_flats_center.csv` summary

Understanding the Output

Console Output

The program displays a formatted table with the following columns:

Column	Description
Image	FITS filename
Object	Target name from FITS header
ExpTime	Exposure time in seconds
Filter	Filter used (if recorded)

Column	Description
Date	Observation date
UT	Universal Time of observation
Julian Date	JD timestamp
Saturated	Number of saturated pixels (>65000 ADU)
TotalPix	Total pixels analyzed
% Saturated	Percentage of saturated pixels
Mean	Mean ADU value in analyzed region

Interpreting Results

Good (No Saturation):

```
NGC7331_001.fits    NGC 7331    300    R    2025-08-20    02:15:32
2460542.594    0    40000    0.00    18543.21
```

- 0% saturation is ideal for science images

Warning (Minor Saturation):

```
NGC7331_002.fits    NGC 7331    300    R    2025-08-20    02:21:15
2460542.598    85    40000    0.21    31420.55
```

- <1% saturation might be acceptable depending on your science goals
- Check if saturated pixels are on your target or just hot pixels

Bad (Significant Saturation):

```
NGC7331_003.fits    NGC 7331    600    R    2025-08-20    02:35:42
2460542.608    4250    40000    10.63    42350.12
```

- 1% saturation usually indicates overexposure
- Image likely unusable for photometry

CSV Output

The program automatically saves results to a CSV file named `{directory}_{mode}.csv`, making it easy to:

- Import into spreadsheets for further analysis
- Track saturation trends across your observing run
- Archive quality control data

Practical Tips

1. **For Galaxy Observations:** Use `-target` mode to focus on the galaxy core, which is most likely to saturate
2. **For Stellar Photometry:** Even a few saturated pixels on your target star will ruin photometric measurements
3. **For Flat Fields:** Aim for mean values around 20,000-30,000 ADU with 0% saturation when using `-center` mode
4. **Quick Quality Check:** Run the program immediately after observations to catch saturation issues while you can still re-observe
5. **Exposure Time Planning:** If multiple images show near-saturation (mean >50,000 ADU), consider reducing exposure times

Example Workflow

```
# After a night of observing NGC 7331
cd /data/2025-08-20/

# Check if the galaxy saturated in any frames
python exposecheck.py NGC7331 -target

# If saturation found, check full frames to see extent
python exposecheck.py NGC7331 -all

# Verify your flat fields are good
python exposecheck.py calibration/flats -center

# Check your standard star observations
python exposecheck.py standards/BD+25_4655 -target
```

Troubleshooting

"ERROR: not a 2D image"

- File might be a data cube or corrupted
- Check with: `fitsinfo filename.fits`

No files processed

- Ensure files have .fits, .fit, or .fts extensions
- Check directory path is correct

Very high saturation in all images

- Verify your camera's actual saturation level (might not be 65,000)
- Check if images are in different units (e.g., electrons vs ADU)